

Endocrine and Metabolic Health

Hormonal regulation, glucose and energy metabolism, adipose biology, mineral balance, growth, and closed-loop measurement

Volume 03D of the Health Atlas Research Suite

Purpose

Represent endocrine and metabolic health as coupled feedback, flux, reserve and life-course systems, and connect laboratory, continuous-sensor, imaging and device measurements to clinically meaningful claims.

Evidence, guidance, and standards cutoff: 13 August 2026

August 2026

Document status and intended use

This is a technical evidence synthesis and research-engineering reference, not individualized medical advice, a substitute for clinical evaluation, or authorization to use an experimental device on people. It distinguishes established knowledge from uncertain inference and frontier research. Diagnostic and therapeutic decisions require qualified professionals, appropriate reference standards, and applicable ethics and regulatory review.

Suite relationship

This volume is read with the Atlas of Human Health (master taxonomy), the Library Guide and Cross-walk, and the other domain volumes identified in the cross-reference chapter. Volume 03A, Cardiovascular Health and Measurement Engineering, remains the reference model for the suite.

Evidence boundary

Evidence and standards are current through 13 August 2026. This volume does not provide medication doses, insulin adjustment instructions, diagnostic cutoffs divorced from laboratory method/context, or authorization to build a human insulin-delivery or invasive biosensor system. Reproductive medicine is expanded in Volume 03M; maternal/fetal physiology in 03N; cardiovascular and renal complications in 03A and 03F.

How to use this volume

Endocrine health is not a list of gland values, and metabolic health is not a single weight or glucose threshold. Hormones coordinate time, growth, stress, reproduction, energy, temperature, water, minerals and adaptation across organs. Their concentrations are pulsatile, circadian, state-dependent and influenced by binding, clearance, illness and medication. A complete profile therefore combines phenotype and trajectory with appropriately timed laboratory data, dynamic testing when needed, glucose and energy patterns, body composition and function, cardiovascular-kidney-liver consequences, sleep, nutrition, medication systems, social context and treatment burden. Devices such as CGM and automated insulin delivery are control systems whose safety depends on sensor, algorithm, actuator and human interaction.

Questions this volume is designed to answer

1. How do endocrine feedback and metabolic flux differ from static laboratory values?
2. How do genes, development, food systems, sleep, stress, medications and environment interact?
3. What are the major endocrine/metabolic failure modes and organ consequences?
4. How should laboratory discordance and dynamic testing be reasoned through?
5. What can CGM, pumps, DXA, bioimpedance, calorimetry and multi-omics actually measure?
6. How should closed-loop systems fail safely?
7. Which prevention and treatment strategies improve patient-important outcomes and where are long-term uncertainties?

Scope

- Hormone signaling, feedback axes, glucose, lipid, energy, thyroid, adrenal, mineral, growth and reproductive metabolism.
- Development, circadian timing, stress, nutrition, body composition, aging and organ coupling.
- Diabetes, obesity, thyroid, adrenal, pituitary, bone/mineral, reproductive-endocrine, rare metabolic and malnutrition phenotypes.
- Laboratory assays, dynamic tests, CGM, pumps, body composition, calorimetry, imaging and multi-omics.
- Closed-loop engineering, assay interference, calibration, algorithm and human-factors validation.
- Prevention, nutrition/activity support, pharmacology, surgery, hormone replacement/suppression and precision therapy.

Reading conventions

- **Established:** replicated mechanisms, mature measurement standards, or interventions supported by convergent evidence.
- **Context-limited:** useful under defined populations, settings, reference methods, or intended uses; transportability must be demonstrated.
- **Frontier:** biologically plausible or technically promising but not yet validated for broad clinical claims.
- **Measurement claim:** always interpreted as an inference from a signal chain rather than a direct view of the latent biological state.

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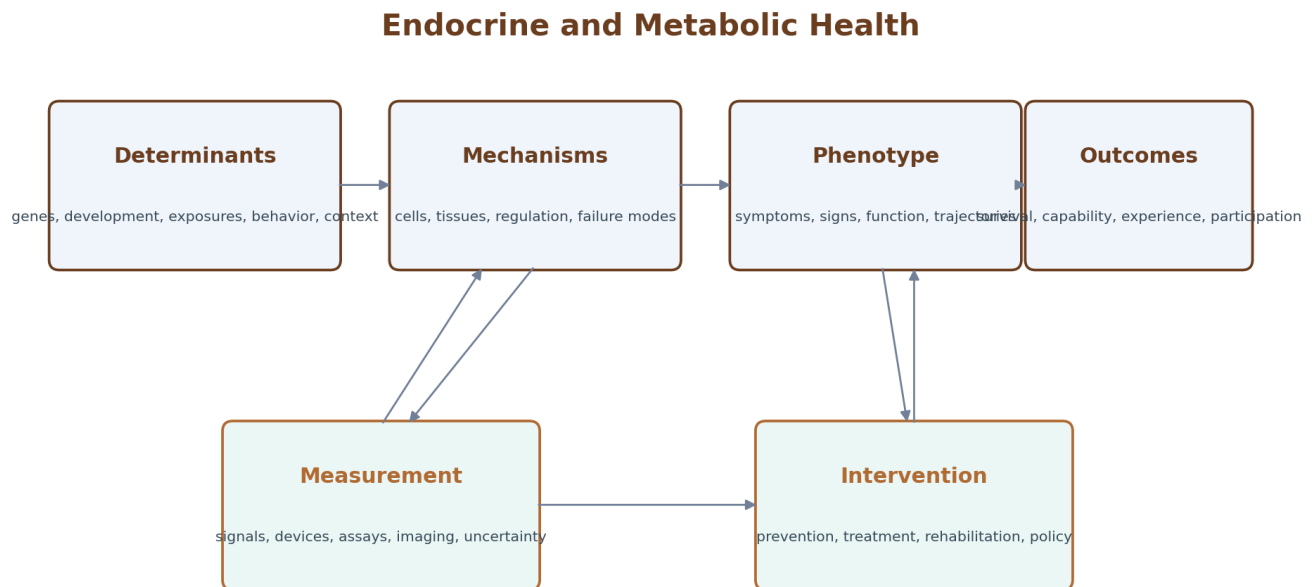
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Executive synthesis

Endocrine health is not a list of gland values, and metabolic health is not a single weight or glucose threshold. Hormones coordinate time, growth, stress, reproduction, energy, temperature, water, minerals and adaptation across organs. Their concentrations are pulsatile, circadian, state-dependent and influenced by binding, clearance, illness and medication. A complete profile therefore combines phenotype and trajectory with appropriately timed laboratory data, dynamic testing when needed, glucose and energy patterns, body composition and function, cardiovascular-kidney-liver consequences, sleep, nutrition, medication systems, social context and treatment burden. Devices such as CGM and automated insulin delivery are control systems whose safety depends on sensor, algorithm, actuator and human interaction.

Central organizing principle

Represent endocrine and metabolic health as coupled feedback, flux, reserve and life-course systems, and connect laboratory, continuous-sensor, imaging and device measurements to clinically meaningful claims.



A health profile is a causal, temporal, functional, and measurement model - not a single score.

Figure 1.1: The causal and measurement architecture used throughout this volume.

1.1 What a complete profile requires

- A multiscale account from molecules and cells through tissues, organs, whole-person function, and context.

- A longitudinal account of baseline, reserve under challenge, variability, compensation, injury, recovery, and irreversible change.
- Separation of causal determinants from correlates, biomarkers, manifestations, and consequences.
- Explicit representation of measurement error, missingness, artifact, reference methods, population drift, and clinical actionability.
- Integration with other atlas domains rather than treating one specialty as a closed system.

Safety boundary

Severe hypoglycemia, diabetic ketoacidosis, hyperosmolar crisis, adrenal crisis, thyroid storm, severe electrolyte disturbance or acute pituitary compression are emergencies. Experimental biosensors, insulin pumps and closed-loop controllers can cause rapid serious harm if sensor, software or delivery fails; human use requires regulated design and clinical oversight.

Historical development and changing models

The history of a domain is not decorative. It explains why present-day categories, instruments, thresholds, specialties, and blind spots look the way they do. Technical advances often changed the object of medicine itself: what could be seen, counted, localized, or treated became the preferred unit of explanation.

Period	Development	What it made possible	Residual limitation
Ancient to 1800s	Diabetes, goiter, wasting, obesity, growth disorders and reproductive changes were recognized clinically.	Established syndromes of altered energy, growth and gland function.	Internal secretions, feedback and molecular causes were unknown.
1800s	Experimental gland removal/replacement and organ extracts demonstrated internal chemical regulation.	Created the concept of endocrine secretion and causal hormone deficiency.	Extracts were crude and feedback relationships remained unclear.
1900-1950	Hormones were isolated; insulin transformed diabetes; radioisotopes and basal metabolic measures expanded thyroid science.	Made lethal deficiencies treatable and metabolism quantifiable.	Dosing, purity, monitoring and chronic complications remained major problems.
1950-1980	Radioimmunoassay, feedback-axis testing, chromatography and modern epidemiology enabled low-concentration measurement.	Separated primary from central disorders and established dynamic testing.	Immunoassays introduced cross-reactivity, matrix and calibration problems still relevant today.
1980-2005	Home glucose meters, HbA1c standardization, pumps, recombinant hormones, DXA and molecular genetics matured.	Enabled long-term self-management, complication prevention and inherited diagnosis.	Access, user burden and population-specific reference limits constrained benefit.
2005-2020	Continuous glucose monitoring, incretin therapies, bariatric/metabolic surgery, high-resolution imaging and multi-omics expanded.	Improved dynamic measurement and treatment of diabetes/obesity; clarified heterogeneous endocrine tumors and rare disease.	Surrogate improvements do not always translate to outcomes; data overload and commercialization increased.
2020-2026	Automated insulin delivery, newer obesity pharmacotherapy, digital therapeutics and precision molecular treatment accelerated.	Produced large benefits for selected people and more closed-loop control.	Long-term durability, lean-mass effects, discontinuation, cost, supply and equity remain critical.

Continued on next page

Period	Development	What it made possible	Residual limitation
Current systems view	Endocrine and metabolic health is understood as coupled energy, growth, stress, electrolyte, bone, reproductive and circadian regulation across the life course.	Integrates organs traditionally separated into diabetes, obesity, thyroid, adrenal, pituitary, bone and reproductive specialties.	Clinical care and research still overuse isolated thresholds and underrepresent food systems, sleep, medications and social determinants.

2.0.1 Diabetes, goiter, wasting, obesity, growth disorders and reproductive changes were recognized clinically.

Context. Endocrine disease was visible long before hormones were measurable.

Scientific consequence. Established syndromes of altered energy, growth and gland function.

What remained unresolved. Internal secretions, feedback and molecular causes were unknown.

Evidence anchors: [3,5]

2.0.2 Experimental gland removal/replacement and organ extracts demonstrated internal chemical regulation.

Context. Physiology shifted from anatomy alone to circulating signals.

Scientific consequence. Created the concept of endocrine secretion and causal hormone deficiency.

What remained unresolved. Extracts were crude and feedback relationships remained unclear.

Evidence anchors: [3,4]

2.0.3 Hormones were isolated; insulin transformed diabetes; radioisotopes and basal metabolic measures expanded thyroid science.

Context. Therapeutics and measurement co-evolved.

Scientific consequence. Made lethal deficiencies treatable and metabolism quantifiable.

What remained unresolved. Dosing, purity, monitoring and chronic complications remained major problems.

Evidence anchors: [6,7]

2.0.4 Radioimmunoassay, feedback-axis testing, chromatography and modern epidemiology enabled low-concentration measurement.

Context. Laboratory endocrinology became central to diagnosis.

Scientific consequence. Separated primary from central disorders and established dynamic testing.

What remained unresolved. Immunoassays introduced cross-reactivity, matrix and calibration problems still relevant today.

Evidence anchors: [7,26]

2.0.5 Home glucose meters, HbA1c standardization, pumps, recombinant hormones, DXA and molecular genetics matured.

Context. Chronic endocrine disease became a device-mediated care domain.

Scientific consequence. Enabled long-term self-management, complication prevention and inherited diagnosis.

What remained unresolved. Access, user burden and population-specific reference limits constrained benefit.

Evidence anchors: [6,21]

2.0.6 Continuous glucose monitoring, incretin therapies, bariatric/metabolic surgery, high-resolution imaging and multi-omics expanded.

Context. Metabolism moved from static fasting values toward continuous dynamics.

Scientific consequence. Improved dynamic measurement and treatment of diabetes/obesity; clarified heterogeneous endocrine tumors and rare disease.

What remained unresolved. Surrogate improvements do not always translate to outcomes; data overload and commercialization increased.

Evidence anchors: [6,8,22]

2.0.7 Automated insulin delivery, newer obesity pharmacotherapy, digital therapeutics and precision molecular treatment accelerated.

Context. Endocrine engineering now includes sensors, algorithms, drugs and behavior as one system.

Scientific consequence. Produced large benefits for selected people and more closed-loop control.

What remained unresolved. Long-term durability, lean-mass effects, discontinuation, cost, supply and equity remain critical.

Evidence anchors: [6,8,24]

2.0.8 Endocrine and metabolic health is understood as coupled energy, growth, stress, electrolyte, bone, reproductive and circadian regulation across the life course.

Context. A normal laboratory value can coexist with low reserve or treatment burden.

Scientific consequence. Integrates organs traditionally separated into diabetes, obesity, thyroid, adrenal, pituitary, bone and reproductive specialties.

What remained unresolved. Clinical care and research still overuse isolated thresholds and underrepresent food systems, sleep, medications and social determinants.

Evidence anchors: ^[15,16,17]

Normal foundations: structure, physiology, and regulation

Normal physiology is represented here as regulated ranges and adaptive control, not as one fixed number. Many variables are intentionally dynamic, tissue-specific, state-dependent, and coupled to other systems. Disease can arise from primary damage, maladaptive compensation, loss of reserve, mistimed regulation, or mismatch between organism and environment.

3.0.1 Hormone signaling and feedback

Core principle. Hormones act through cell-surface or intracellular receptors with amplification, feedback and tissue-specific response.

Control architecture. Hypothalamic-pituitary axes, autonomic input, binding proteins, metabolism and receptor sensitivity regulate concentration and effect.

Failure modes. Deficiency, excess, resistance, ectopic production, altered binding or assay interference creates discordant phenotypes.

Observable consequences. Timed hormone concentrations, dynamic tests, binding proteins, downstream physiology and symptoms.

Cross-system coupling. Liver, kidney, nutrition, drugs, pregnancy, sleep and acute illness alter kinetics and interpretation.

Evidence anchors: [3,4,30]

3.0.2 Glucose homeostasis

Core principle. Insulin, glucagon, incretins, liver, muscle, adipose, kidney and brain coordinate glucose supply and disposal across fasting and feeding.

Control architecture. Beta-cell sensing, hepatic production, peripheral uptake, counterregulation and autonomic/hormonal responses maintain range.

Failure modes. Insulin resistance, beta-cell failure, autoimmune destruction, drug effects or endocrine excess produces dysglycemia.

Observable consequences. Plasma glucose, HbA1c, CGM, tolerance testing, insulin/C-peptide in selected contexts and complications.

Cross-system coupling. Food, activity, sleep, stress, infection, kidney/liver function and medications determine dynamics.

Evidence anchors: [3,4,6]

3.0.3 Energy balance and adipose biology

Core principle. Body energy stores reflect intake, absorption, expenditure and partitioning; adipose tissue is an endocrine and immune organ.

Control architecture. Hypothalamic sensing, gut peptides, reward, thermogenesis, muscle, sleep and environment regulate behavior and expenditure.

Failure modes. Chronic positive balance, adipose dysfunction, ectopic fat, inflammation or lipodystrophy produces metabolic disease.

Observable consequences. Weight trajectory, waist, body composition, energy intake/expenditure, metabolic markers and function.

Cross-system coupling. Food systems, medications, stress, poverty, built environment, genetics and life stage interact.

Evidence anchors: [8,15,16]

3.0.4 Lipid transport and storage

Core principle. Lipoproteins transport triglyceride and cholesterol among intestine, liver and tissues; cells regulate synthesis, uptake and efflux.

Control architecture. Apolipoproteins, lipases, receptors, bile pathways, hormones and nutrition control flux.

Failure modes. Dysregulated production/clearance causes atherogenic particles, pancreatitis risk, fatty liver or storage disorders.

Observable consequences. Lipid panel, apoB, Lp(a), triglycerides, genetic tests selectively and tissue consequences.

Cross-system coupling. Cardiovascular, hepatic, renal, endocrine and genetic systems converge.

Evidence anchors: [2,4,21]

3.0.5 Thyroid regulation and thermogenesis

Core principle. Thyroid hormone sets transcriptional programs affecting metabolic rate, growth, heart, brain, muscle and temperature.

Control architecture. TRH-TSH-thyroid feedback, iodine, deiodinases, binding proteins and illness regulate availability.

Failure modes. Autoimmune, nodular, central, drug, iodine or congenital disorders cause excess or deficiency.

Observable consequences. TSH and free hormones interpreted by context, antibodies, imaging and function.

Cross-system coupling. Pregnancy, age, acute illness, medications and pituitary disease alter reference and consequences.

Evidence anchors: [4,7,14]

3.0.6 Stress and adrenal axes

Core principle. Cortisol, catecholamines, aldosterone and adrenal androgens coordinate stress, vascular tone, glucose, salt and immune responses.

Control architecture. Circadian and pulsatile ACTH, renin-angiotensin-potassium, autonomic input and feedback regulate secretion.

Failure modes. Primary/central failure, autonomous excess, tumors, enzyme defects or chronic exogenous steroid exposure disrupts adaptation.

Observable consequences. Timed and dynamic hormones, electrolytes, pressure, imaging and clinical phenotype.

Cross-system coupling. Cardiovascular, renal, immune, psychosocial and sleep systems strongly couple.

Evidence anchors: [2,7,15]

3.0.7 Calcium, phosphate and bone remodeling

Core principle. Bone, kidney, gut and parathyroid/vitamin D axes regulate mineral availability and skeletal strength.

Control architecture. PTH, vitamin D, FGF23, sex steroids, mechanical loading and renal handling coordinate flux.

Failure modes. Deficiency, excess, CKD, malabsorption, immobility, drugs or endocrine disease causes fragility and ectopic calcification.

Observable consequences. Calcium/phosphate/PTH/vitamin D, DXA, fracture history, imaging and bone turnover markers selectively.

Cross-system coupling. Renal, GI, reproductive, musculoskeletal and aging systems are inseparable.

Evidence anchors: [9,10,11]

3.0.8 Growth and development

Core principle. Growth hormone/IGF, thyroid, sex steroids, nutrition and developmental timing regulate linear growth and tissue maturation.

Control architecture. Pulsatile secretion, sleep, liver response, growth plates, genetics and puberty coordinate trajectory.

Failure modes. Deficiency/excess, chronic disease, malnutrition, genetic syndromes or altered puberty changes growth and body composition.

Observable consequences. Longitudinal growth velocity, pubertal stage, bone age, targeted hormones/genetics and function.

Cross-system coupling. Maternal-fetal health, nutrition, chronic disease and psychosocial context shape growth.

Evidence anchors: [7,17,18]

3.0.9 Reproductive endocrine integration

Core principle. Hypothalamic-pituitary-gonadal signaling coordinates puberty, gametogenesis, cycles, pregnancy capability, sexual function and bone.

Control architecture. Pulsatile GnRH, gonadotropins, gonadal hormones, energy availability and feedback regulate function.

Failure modes. Primary/central gonadal failure, hyperandrogenism, energy deficit, tumors, genetics or medications impair function.

Observable consequences. History, cycles/puberty, timed hormones, imaging, fertility testing and bone/metabolic consequences.

Cross-system coupling. Reproductive, psychological, nutritional and life-course domains must be integrated.

Evidence anchors: [7,12,13]

3.0.10 Water, sodium and osmotic regulation

Core principle. Hypothalamic thirst and vasopressin coordinate kidney water handling to protect osmolality and volume.

Control architecture. Osmoreceptors, baroreceptors, collecting-duct response and access to water regulate balance.

Failure modes. Diabetes insipidus, inappropriate antidiuresis, adrenal failure, polydipsia or kidney disease causes sodium/water disorders.

Observable consequences. Serum/urine osmolality and sodium, volume status, urine output and dynamic testing selectively.

Cross-system coupling. Renal, neurologic, cardiovascular and medication effects are central.

Evidence anchors: [4,7,11]

3.0.11 Circadian and meal-timed metabolism

Core principle. Hormones, sleep-wake timing, feeding, light and activity create daily rhythms in glucose, cortisol, thyroid and energy use.

Control architecture. Suprachiasmatic and peripheral clocks, melatonin, behavior and social schedules synchronize tissues.

Failure modes. Shift work, sleep loss, irregular meals, illness and medications produce misalignment and altered risk.

Observable consequences. Sleep/activity/light/meal logs, CGM, timed hormones and function.

Cross-system coupling. Psychosocial, occupational, sleep and environmental systems determine timing.

Evidence anchors: [15,19,20]

3.1 Useful quantitative relationships

These equations are compact models. Their variables require operational definitions, units, boundary conditions, and a statement of when the approximation fails.

Relationship	Expression	Interpretation	Limits
Mass balance	$\Delta S = \text{input} - \text{output}$	Change in a body store equals absorbed/produced input minus utilization/excretion output.	Stores and fluxes are compartmental, time-varying and incompletely measured.
HOMA-IR approximation	$\text{HOMA-IR} = \frac{G_0 I_0}{22.5}$	A fasting glucose-insulin proxy for insulin resistance under specified units and assumptions.	Not a universal individual diagnostic; insulin assays differ and beta-cell failure alters interpretation.
Body mass index	$BMI = \frac{m}{h^2}$	Simple mass-to-height index useful for population screening.	Does not directly measure fat distribution, composition, ethnicity/age differences or health; cutoffs are context tools.
Free hormone concept	$H_{free} \rightleftharpoons H_{bound}$	Only a fraction of many hormones is unbound, while binding proteins buffer and transport total hormone.	Free assays and equilibrium assumptions can fail; tissue uptake and local metabolism matter.
Exponential elimination	$C(t) = C_0 e^{-kt}$	First-order decline approximates many hormone or drug concentrations.	Endogenous pulsatility, saturable clearance, compartments and feedback violate simple decay.

Determinants, heterogeneity, and causal architecture

No universal percentage can divide health into genetics and environment. Contributions vary by outcome, age, population, exposure distribution, measurement scale, and causal estimand. Genetic liability can alter sensitivity to an exposure; exposures can alter gene regulation; both operate through development, behavior, social conditions, and access to prevention or care.

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Genetic architecture and family history	Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within endocrine and metabolic health.	Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.	Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.	Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.
Development and life course	Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.	Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.	Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.	Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.
Nutrition and metabolism	Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.	Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.	Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.	Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.
Physical activity, conditioning, and sedentary behavior	Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.	Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.	Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.	Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Sleep, circadian timing, and recovery	Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.	Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.	History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.	Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.
Stress, emotion, and behavior	Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.	Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.	Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.	Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.
Social, economic, cultural, and care conditions	Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.	Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.	Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.	Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.
Environmental and occupational exposures	Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.	Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.	Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.	Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.
Health care, medication, and iatrogenic effects	Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.	Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.	Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.	Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

4.0.1 Genetic architecture and family history

Mechanistic pathways. Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within endocrine and metabolic health.

Potentially modifiable components. Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.

How to measure it. Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.

Causal and equity cautions. Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.

Evidence anchors: ^[21]

4.0.2 Development and life course

Mechanistic pathways. Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.

Potentially modifiable components. Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.

How to measure it. Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.

Causal and equity cautions. Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.

Evidence anchors: ^[17,18]

4.0.3 Nutrition and metabolism

Mechanistic pathways. Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.

Potentially modifiable components. Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.

How to measure it. Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.

Causal and equity cautions. Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

Evidence anchors: ^[8]

4.0.4 Physical activity, conditioning, and sedentary behavior

Mechanistic pathways. Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.

Potentially modifiable components. Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.

How to measure it. Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.

Causal and equity cautions. Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

Evidence anchors: ^[10]

4.0.5 Sleep, circadian timing, and recovery

Mechanistic pathways. Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.

Potentially modifiable components. Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.

How to measure it. History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.

Causal and equity cautions. Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.

Evidence anchors: ^[15,19]

4.0.6 Stress, emotion, and behavior

Mechanistic pathways. Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.

Potentially modifiable components. Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.

How to measure it. Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.

Causal and equity cautions. Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.

Evidence anchors: ^[15,16]

4.0.7 Social, economic, cultural, and care conditions

Mechanistic pathways. Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.

Potentially modifiable components. Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.

How to measure it. Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.

Causal and equity cautions. Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.

Evidence anchors: [16,34]

4.0.8 Environmental and occupational exposures

Mechanistic pathways. Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.

Potentially modifiable components. Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.

How to measure it. Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.

Causal and equity cautions. Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

Evidence anchors: [20]

4.0.9 Health care, medication, and iatrogenic effects

Mechanistic pathways. Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.

Potentially modifiable components. Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.

How to measure it. Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.

Causal and equity cautions. Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

Evidence anchors: [31,38]

Major disorders, failure modes, and clinically important phenotypes

The profiles below are not a replacement for a diagnostic manual. They are organized to connect mechanism, phenotype, measurement, intervention, and uncertainty. The primary unit is the endocrine axis disorder, metabolic phenotype, crisis, or organ-risk state, but cross-system processes and functional effects are retained.

5.0.1 Type 1 diabetes

Definition and phenotype. Autoimmune or other beta-cell destruction causes absolute insulin deficiency and risk of ketoacidosis.

Pathophysiology. Immune-mediated loss of insulin secretion disrupts glucose uptake and suppressive control of lipolysis/ketogenesis.

Determinants and modifiers. Genetic susceptibility and environmental triggers interact; access to insulin, education, food, devices and care drives outcomes.

Assessment. Glucose/HbA1c, autoantibodies and C-peptide when classification matters; ketones/acid-base when ill; complication and psychosocial screening.

Intervention logic. Lifelong insulin with education, CGM and automated delivery when available, nutrition/activity integration, hypoglycemia prevention and complication risk reduction.

Known limits and open questions. Disease-stage screening and immune delay strategies are evolving; device access and algorithm performance vary.

Suite links. 03C cognition/hypoglycemia; 03F kidney; 03A vascular; 14 devices/pharmacology.

Evidence anchors: [6,21,23]

5.0.2 Type 2 diabetes

Definition and phenotype. Hyperglycemia arising from insulin resistance, beta-cell dysfunction and interacting hepatic, adipose, kidney, gut and brain mechanisms.

Pathophysiology. Progressive mismatch between insulin demand and secretion causes fasting/postprandial dysglycemia and vascular/metabolic injury.

Determinants and modifiers. Genetics, adiposity distribution, food and activity environment, sleep, medications, age, pregnancy history, social conditions and organ disease.

Assessment. Validated glucose/HbA1c criteria with repeat/clinical confirmation as appropriate; phenotype, complications, cardiovascular/kidney risk and treatment burden.

Intervention logic. Lifestyle and social support; individualized glucose-lowering therapy emphasizing cardiovascular/kidney/weight benefits where indicated; monitoring and complication prevention.

Known limits and open questions. Labels conceal heterogeneous beta-cell, resistance and organ phenotypes; remission definitions and long-term durability vary.

Suite links. 03A cardiovascular; 03F renal; 03E liver; 06 social; 11 prevention.

Evidence anchors: [6,8,16]

5.0.3 Prediabetes and dysglycemic risk states

Definition and phenotype. Glucose measures above usual range but below diabetes thresholds, indicating heterogeneous future risk.

Pathophysiology. Early insulin resistance, beta-cell compensation/failure and context-dependent glucose excursions.

Determinants and modifiers. Age, family history, prior gestational diabetes, adiposity, sleep, medication and social/environmental factors.

Assessment. Standardized fasting glucose, HbA1c or tolerance testing; recognize assay, anemia and ethnicity/context limitations; evaluate overall risk.

Intervention logic. Intensive behavior/weight support, treatment of cardiovascular risk and selected medication in high-risk groups.

Known limits and open questions. Thresholds create categories on a continuum; many never progress and some complications reflect shared risk rather than glucose alone.

Suite links. 11 prevention; 03A cardiovascular; 03N gestational diabetes.

Evidence anchors: [6,41]

5.0.4 Obesity and adiposity-related disease

Definition and phenotype. Chronic heterogeneous disease of excess or dysfunctional adiposity that can impair health, function or quality of life.

Pathophysiology. Energy imbalance interacts with neuroendocrine regulation, adipocyte expansion, ectopic fat, inflammation, mechanics and organ-specific susceptibility.

Determinants and modifiers. Genetics, developmental environment, food systems, medications, sleep, stress, activity opportunity, poverty, stigma and built environment.

Assessment. Weight trajectory, BMI as screening, waist/body composition selectively, complications, function, eating/mental health and goals.

Intervention logic. Nonstigmatizing nutrition/activity/behavior support, anti-obesity medication and metabolic surgery according to severity, complications, preference and access; long-term maintenance.

Known limits and open questions. BMI is an imperfect proxy; optimal definitions, drug duration, lean-mass effects, weight cycling and population transport remain debated.

Suite links. 03A/03B/03E/03I organ consequences; 06 psychosocial/stigma; 11 policy.

Evidence anchors: [6,8,34]

5.0.5 Hypoglycemia

Definition and phenotype. Glucose low enough to produce autonomic/neuroglycopenic symptoms or require assistance, commonly treatment-related.

Pathophysiology. Insulin or secretagogue excess, impaired counterregulation, fasting, exercise, alcohol, organ failure or rare endogenous hyperinsulinism reduces brain fuel.

Determinants and modifiers. Diabetes treatment, kidney/liver disease, age, cognition, food insecurity, exercise, pregnancy and medication errors.

Assessment. Confirm glucose during symptoms when possible; classify severity/context; review CGM accuracy and medications; evaluate unexplained episodes with critical samples.

Intervention logic. Immediate carbohydrate or glucagon as appropriate, regimen/technology adjustment, education, awareness restoration and cause treatment.

Known limits and open questions. Interstitial CGM lag and compression artifact can misclassify events; recurrent episodes alter awareness and risk.

Suite links. 03C neurologic; 03F kidney; 12 safety; 14 medication systems.

Evidence anchors: [6,31]

5.0.6 Diabetic ketoacidosis and hyperosmolar crisis

Definition and phenotype. Acute metabolic emergencies caused by severe insulin deficiency and/or extreme hyperglycemia with dehydration and electrolyte disturbance.

Pathophysiology. Lipolysis/ketogenesis and osmotic diuresis produce acidosis, volume depletion and total-body potassium deficit despite variable serum values.

Determinants and modifiers. Infection, missed insulin, new diabetes, drugs, access barriers and device failure.

Assessment. Glucose, ketones, blood gas, electrolytes/osmolality, renal function, precipitant and serial neurologic/hemodynamic assessment.

Intervention logic. Fluids, insulin, potassium/electrolyte replacement and cause treatment with protocolized monitoring.

Known limits and open questions. Definitions and fluid strategies vary by age/context; cerebral edema and overly rapid shifts are rare but serious.

Suite links. 03F renal/electrolyte; 08 infection; 12 acute care.

Evidence anchors: [6,31]

5.0.7 Thyroid dysfunction

Definition and phenotype. Hypothyroidism or hyperthyroidism results from thyroid, pituitary/hypothalamic, autoimmune, nodular, drug, iodine or inflammatory causes.

Pathophysiology. Altered thyroid hormone production, release, conversion, binding or feedback changes metabolic and organ function.

Determinants and modifiers. Autoimmunity, iodine exposure, pregnancy/postpartum, medications, radiation, nodules and pituitary disease.

Assessment. TSH with free hormone interpreted by suspected level and context; antibodies and imaging selectively; repeat unexpected discordance and evaluate assay interference.

Intervention logic. Hormone replacement for deficiency; antithyroid drugs, radioactive iodine or surgery for selected excess; emergency management for severe decompensation.

Known limits and open questions. Subclinical thresholds, symptoms and treatment benefit vary; acute illness and pregnancy require distinct interpretation.

Suite links. 03A rhythm/heart; 03N pregnancy; 05 mood/cognition; 14 pharmacology.

Evidence anchors: [7,14,30]

5.0.8 Adrenal insufficiency

Definition and phenotype. Inadequate cortisol with or without aldosterone due to primary adrenal, central or glucocorticoid-withdrawal causes.

Pathophysiology. Loss of stress hormone and mineralocorticoid support impairs vascular tone, glucose and sodium balance.

Determinants and modifiers. Autoimmune disease, infection, hemorrhage, pituitary disease, surgery, genetics and chronic exogenous steroids.

Assessment. Morning cortisol/ACTH and dynamic testing as appropriate; electrolytes, renin/aldosterone for primary disease; urgent clinical diagnosis in crisis.

Intervention logic. Glucocorticoid replacement and stress dosing; mineralocorticoid in primary disease; emergency parenteral treatment for crisis; education and identification.

Known limits and open questions. Assay and timing thresholds vary; critical illness physiology complicates diagnosis; chronic dose optimization is imperfect.

Suite links. 03A hemodynamics; 03F sodium; 08 autoimmunity/infection; 12 emergency care.

Evidence anchors: ^[7,31]

5.0.9 Cortisol excess and Cushing syndrome

Definition and phenotype. Chronic inappropriate glucocorticoid exposure from medication or endogenous pituitary, adrenal or ectopic production.

Pathophysiology. Cortisol excess promotes insulin resistance, catabolism, immune suppression, hypertension, thrombosis, bone loss and neuropsychiatric effects.

Determinants and modifiers. Exogenous steroids are common; endogenous tumors are rare; alcohol, depression and severe obesity can mimic biochemical features.

Assessment. Establish clinical probability and exclude exogenous exposure; use validated screening tests and repeat/confirm before localization; consider cyclic disease and assay interference.

Intervention logic. Reduce exogenous exposure safely; surgery or cause-directed drugs/radiation; treat cardiovascular, metabolic, bone, infection and psychiatric consequences.

Known limits and open questions. False positives are common in low-pretest populations; localization tests require expert integration.

Suite links. 03A cardiovascular; 03G thrombosis; 03I bone/muscle; 05 mental health.

Evidence anchors: ^[7,41]

5.0.10 Primary aldosteronism and endocrine hypertension

Definition and phenotype. Autonomous aldosterone or other endocrine pathways cause potentially treatable hypertension and organ injury.

Pathophysiology. Sodium retention and potassium loss with maladaptive cardiovascular/renal signaling; catecholamine or cortisol excess are other mechanisms.

Determinants and modifiers. Adrenal adenoma/hyperplasia, familial forms and rare tumors; detection depends on who is screened.

Assessment. Medication-aware renin/aldosterone testing and confirmation/localization as appropriate; electrolytes, imaging and adrenal venous sampling in selected surgical candidates.

Intervention logic. Mineralocorticoid blockade or adrenalectomy by subtype; treat pressure and organ risk.

Known limits and open questions. Screening thresholds and medication washout are complex; imaging alone does not establish laterality.

Suite links. 03A hypertension; 03F kidney; 14 pharmacology/procedure.

Evidence anchors: ^[2,7]

5.0.11 Pituitary and hypothalamic disorders

Definition and phenotype. Tumor, inflammation, injury, genetics or treatment can cause hormone excess, deficiency, mass effect and water-balance disorders.

Pathophysiology. Altered secretion of one or multiple pituitary axes, stalk effects, compression and hypothalamic dysfunction.

Determinants and modifiers. Adenomas, surgery/radiation, traumatic brain injury, immune/infiltrative disease, postpartum injury and genetic syndromes.

Assessment. Visual and neurologic symptoms, axis-specific timed/dynamic testing, MRI and medication review; evaluate all axes and emergencies.

Intervention logic. Hormone replacement, surgery, targeted drugs or radiation, visual/neurologic management and long-term surveillance.

Known limits and open questions. Incidental lesions are common; hormone assays and dynamic thresholds vary; treatment can trade excess for deficiency.

Suite links. 03C nervous; 03K vision; 03N pregnancy; 14 surgery/pharmacology.

Evidence anchors: ^[7]

5.0.12 Polycystic ovary syndrome

Definition and phenotype. A heterogeneous reproductive-metabolic syndrome involving ovulatory dysfunction and androgen excess after exclusion of important mimics.

Pathophysiology. Ovarian/adrenal androgen production, insulin resistance, follicular development and neuroendocrine regulation interact.

Determinants and modifiers. Genetics, adiposity, developmental factors and environment influence expression; stigma and access affect outcome.

Assessment. Age-appropriate clinical/biochemical criteria, cycle history, pregnancy exclusion, mimic screening and metabolic/mental health assessment; imaging is context-specific.

Intervention logic. Goals-based cycle/endometrial protection, fertility care, acne/hirsutism treatment, metabolic risk reduction and psychosocial support.

Known limits and open questions. Definitions differ across life stages and populations; ovarian morphology alone is not diagnostic; long-term risk estimates vary.

Suite links. 03M reproductive; 05 mental health; 03A metabolic vascular risk; 14 therapeutics.

Evidence anchors: [7,12,13]

5.0.13 Osteoporosis and metabolic bone disease

Definition and phenotype. Reduced bone strength raises fragility-fracture risk due to aging, hormone deficiency, chronic disease, medications or mineral disorders.

Pathophysiology. Imbalanced remodeling, microarchitectural deterioration, low peak bone mass and falls interact.

Determinants and modifiers. Age, sex-steroid loss, low weight, immobility, steroids, kidney/GI/endocrine disease, smoking, alcohol and genetics.

Assessment. Fracture history, clinical risk, DXA with quality control, secondary-cause labs and vertebral imaging selectively; falls/function assessment.

Intervention logic. Exercise/fall prevention, adequate nutrition, cause treatment and antiresorptive/anabolic medication by absolute fracture risk; treatment sequencing and monitoring.

Known limits and open questions. DXA is areal density not complete strength; rare adverse effects and rebound after some drugs require planning.

Suite links. 03I musculoskeletal; 03F renal; 03E malabsorption; 07 aging.

Evidence anchors: [7,9,10]

5.0.14 Hyperparathyroidism, hypoparathyroidism and mineral disorders

Definition and phenotype. Disordered PTH signaling causes abnormal calcium/phosphate, bone, kidney and neuromuscular effects.

Pathophysiology. Autonomous gland secretion, gland failure, vitamin D/renal feedback or receptor resistance changes mineral flux.

Determinants and modifiers. Adenoma/hyperplasia, surgery, autoimmune/genetic disease, kidney disease, vitamin D deficiency and medications.

Assessment. Corrected or ionized calcium, phosphate, PTH, vitamin D, renal function, urine calcium and imaging only after biochemical diagnosis when needed for procedure.

Intervention logic. Surgery for qualifying primary disease; calcium/vitamin D or PTH analog strategies for deficiency; manage renal secondary disease and complications.

Known limits and open questions. Reference intervals and albumin correction have limits; normocalcemic categories and treatment thresholds remain debated.

Suite links. 03F renal; 03I bone; 03C neuromuscular; 14 procedure/pharmacology.

Evidence anchors: ^[7,11]

5.0.15 Inborn errors, mitochondrial and storage metabolism

Definition and phenotype. Genetic defects in enzymes, transporters, organelles or cofactors cause episodic or progressive multisystem disease.

Pathophysiology. Blocked pathways produce substrate accumulation, product deficiency, toxic intermediates or energy failure.

Determinants and modifiers. Genotype and residual activity interact with diet, infection, fasting, drugs and developmental stage.

Assessment. Newborn screening, biochemical pattern, genetics/functional confirmation, organ-specific monitoring and emergency plans.

Intervention logic. Avoid catabolism/toxic substrates, replace products/cofactors, remove metabolites, enzyme/substrate/gene therapies for selected disorders, multidisciplinary care.

Known limits and open questions. Variants and biochemical findings can be ambiguous; newborn detection does not guarantee effective therapy or equal access.

Suite links. 02 molecular biology; 07 development; 14 advanced therapy; all organ volumes.

Evidence anchors: ^[21,22,32]

5.0.16 Malnutrition, sarcopenia and relative energy deficiency

Definition and phenotype. Insufficient energy, protein or micronutrient availability impairs body composition, immunity, endocrine function, growth and performance.

Pathophysiology. Negative balance and inflammatory catabolism reduce muscle and organ reserve; adaptive endocrine suppression may occur.

Determinants and modifiers. Food insecurity, illness, malabsorption, eating disorder, high training load, aging, disability and medications.

Assessment. Weight/growth trajectory, diet and food access, physical exam, function and targeted labs; body composition selectively; diagnose cause rather than one marker.

Intervention logic. Food and social intervention, cause treatment, structured nutrition and resistance/functional rehabilitation; cautious refeeding when severe.

Known limits and open questions. Albumin is not a standalone nutrition measure; body composition methods differ; ideal targets vary by disease and goals.

Suite links. 03E digestion; 03I muscle; 05 eating behavior; 06/11 social systems.

Evidence anchors: ^[10,16,33]

Closed-loop metabolic systems

Closed-loop systems join a sensor, state estimator, controller, actuator and human. Their safety depends on explicit failure handling because physiological delay, meals, exercise, illness, sensor error and infusion failure can all drive the loop outside assumptions.

6.1 Control-system anatomy

Element	Metabolic example	Critical failure
Measurand	Interstitial glucose.	Lag or compression artifact misrepresents blood glucose.
Estimator	Filtered glucose and rate of change.	Smoothing hides rapid transition or uses stale data.
Controller	Insulin dosing rule/model predictive control.	Unannounced meal/exercise or parameter drift causes over/underdelivery.
Actuator	Pump and infusion set.	Occlusion, leakage, dislodgement or wrong concentration.
Human interface	Meal input, alarms, override and rescue plan.	Alarm fatigue, misunderstanding, inaccessible design or no fallback.

Selected evidence anchors: [6,24,27]

6.2 Required fault tests

- Sensor dropout, bias, compression lows and rapid glucose change.
- Infusion occlusion, partial delivery, empty reservoir and site absorption variability.
- Clock mismatch, Bluetooth interruption, phone/OS update and battery depletion.
- Incorrect carbohydrate entry, exercise, alcohol, vomiting and intercurrent illness.
- Safe mode, manual transition, ketone guidance and emergency contact pathways.
- Cybersecurity compromise and unauthorized setting changes.

Selected evidence anchors: [25,29,31]

Interpreting endocrine laboratory discordance

Endocrine values are conditioned on time, posture, fasting, cycle/pregnancy, acute illness, drugs, binding proteins and assay method. An unexpected result should be replicated and explained before it is converted into a diagnosis or imaging cascade.

7.1 Discordance checklist

- Confirm identity, units, timing, specimen and reference interval.
- Review biotin, supplements, antibodies, binding-protein changes and medications.
- Ask whether acute illness or altered sleep/circadian timing invalidates the usual interpretation.
- Compare upstream and downstream axis values rather than one analyte.
- Use an alternate assay or mass spectrometry when interference is plausible.
- Perform dynamic testing only when pretest probability and safety justify it.
- Do not use imaging to diagnose biochemical hormone excess.

Selected evidence anchors: [7,28,41]

A complete metabolic profile

The profile should separate risk, current organ injury, daily dynamics, reserve and treatment burden. A single composite metabolic age or score is not a substitute for these interpretable components.

8.1 Core layers

- Anthropometric and growth trajectory with body composition only when it changes decisions.
- Glucose dynamics, severe hypoglycemia and treatment-system reliability.
- Blood pressure, atherogenic particles and cardiovascular history.
- Kidney filtration/albuminuria, liver phenotype and sleep-disordered breathing.
- Strength, mobility, bone and nutrition status.
- Medication and endocrine disruptor review, reproductive life stage and pregnancy history.
- Food access, work schedule, stress, activity opportunity, culture, mental health and goals.

Selected evidence anchors: [2,6,11,35]

Assessment, clinical reasoning, and longitudinal profiling

Assessment is a staged inference process. The right test depends on the pretest question, urgency, expected change in management, burden, and the consequences of false reassurance or false alarm. A complete profile combines history, examination, validated measurements, function, trajectories, and context.

Stage	Purpose	Components	Output	Common failure
1. Question and urgency	Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.	Chief concern and goals; red flags; prior probability; relevant context and current endocrine and metabolic health risks.	Urgent action, focused differential, or a prespecified measurement plan.	Testing without a question creates incidental findings and false precision.
2. History and lived phenotype	Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.	Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.	A longitudinal problem representation with hypotheses and missing information.	Administrative codes and checklists can erase chronology, context, and patient priorities.
3. Examination and basic measurements	Test hypotheses using standardized observation and low-burden validated measures.	Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.	Localization, severity estimate, safety screen, and indication for further testing.	Nonstandard technique and unblinded interpretation create systematic error.
4. Targeted laboratory, imaging, and physiology	Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.	Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.	Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.	Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.
5. Function and challenge testing	Measure reserve, adaptation, and real-world impact not visible at rest.	Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.	Capacity, symptom coupling, recovery, and safety limits.	Challenge protocols may not match daily demands and can be unsafe without screening.
6. Longitudinal integration	Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.	Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.	Trajectory, response, relapse, and need for escalation or de-escalation.	Regression to the mean and changing devices can masquerade as improvement or decline.

9.0.1 1. Question and urgency

Purpose. Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.

Core components. Chief concern and goals; red flags; prior probability; relevant context and current endocrine and metabolic health risks.

Interpretation. Urgent action, focused differential, or a prespecified measurement plan.

Failure modes. Testing without a question creates incidental findings and false precision.

Evidence anchors: ^[41]

9.0.2 2. History and lived phenotype

Purpose. Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.

Core components. Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.

Interpretation. A longitudinal problem representation with hypotheses and missing information.

Failure modes. Administrative codes and checklists can erase chronology, context, and patient priorities.

9.0.3 3. Examination and basic measurements

Purpose. Test hypotheses using standardized observation and low-burden validated measures.

Core components. Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.

Interpretation. Localization, severity estimate, safety screen, and indication for further testing.

Failure modes. Nonstandard technique and unblinded interpretation create systematic error.

Evidence anchors: ^[4,5]

9.0.4 4. Targeted laboratory, imaging, and physiology

Purpose. Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.

Core components. Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.

Interpretation. Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.

Failure modes. Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.

Evidence anchors: ^[41]

9.0.5 5. Function and challenge testing

Purpose. Measure reserve, adaptation, and real-world impact not visible at rest.

Core components. Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.

Interpretation. Capacity, symptom coupling, recovery, and safety limits.

Failure modes. Challenge protocols may not match daily demands and can be unsafe without screening.

9.0.6 6. Longitudinal integration

Purpose. Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.

Core components. Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.

Interpretation. Trajectory, response, relapse, and need for escalation or de-escalation.

Failure modes. Regression to the mean and changing devices can masquerade as improvement or decline.

Measurement and device landscape

A device does not measure health directly. It senses a physical or chemical proxy, transforms it, samples it, rejects or retains artifact, and maps the result to a claim. The claim may concern detection, diagnosis, severity, risk, response, or function; each requires a different validation design.

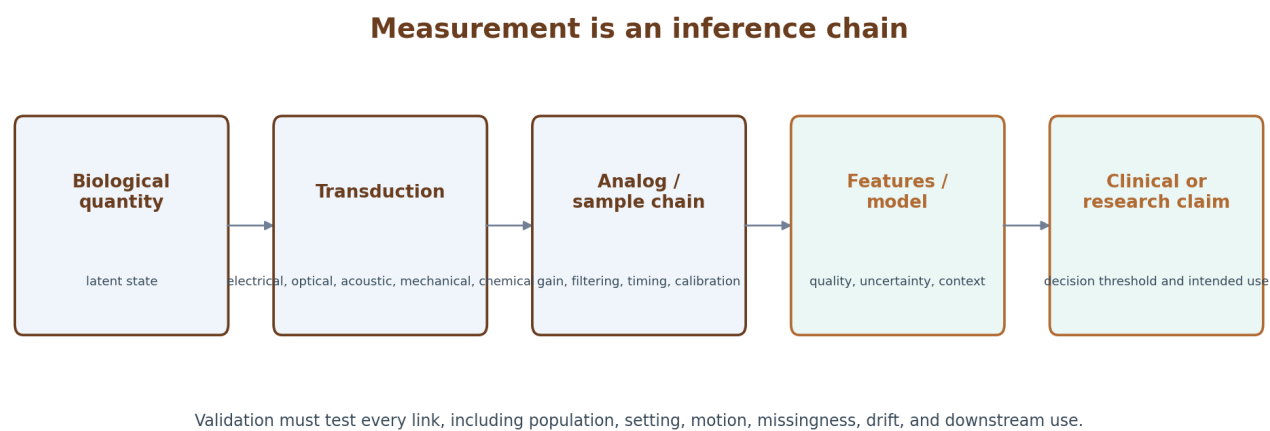


Figure 10.1: Generic inference chain from latent biology to a clinical or research claim.

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Clinical history, anthropometry and growth charts	Noninvasive	Symptoms, medication/exposure history, weight, height, waist, growth velocity and pubertal stage.	Trajectory, phenotype, nutritional and functional context.	Low cost and often more informative than one laboratory value.	Technique, edema, clothing, body composition, reference population, stigma and recall.
Laboratory immunoassays	Minimally invasive specimen	Antibody-based detection of hormones, binding proteins, antibodies or peptides.	Concentration interpreted by timing, reference and axis.	Sensitive, scalable and central to endocrine diagnosis.	Cross-reactivity, heterophile/biotin interference, hook effect, matrix and platform non-comparability.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Mass spectrometry and chromatography	Minimally invasive specimen	Separation and mass/charge detection of steroids, metabolites, amino acids and drugs.	Specific analyte panels, metabolite patterns and confirmatory measurements.	Higher analytical specificity and multiplexing.	Complex preanalytics, calibration, isomers, cost and laboratory-developed method variability.
Plasma glucose meter	Minimally invasive capillary	Electrochemical enzymatic strip measures capillary glucose.	Point glucose for self-management and acute decisions.	Fast, portable and actionable.	Strip/storage, hematocrit, perfusion, contamination, interfering substances and user technique; capillary may differ in shock.
Continuous glucose monitoring	Minimally invasive interstitial sensor	Enzymatic or optical sensing estimates interstitial glucose continuously.	Glucose trace, time in range, variability, alerts and automated-delivery input.	Reveals dynamics and supports closed-loop control.	Physiologic lag, compression, calibration/model drift, sensor failure, medication interference, alert burden and access.
Insulin pump and automated delivery telemetry	Minimally invasive therapeutic device	Programmed subcutaneous insulin delivery integrated with CGM/control algorithm.	Delivered dose, basal/bolus history, algorithm state and device alarms.	Precise adjustable delivery and reduced burden for selected users.	Infusion failure can rapidly cause ketoacidosis; cybersecurity, interface, user input and algorithm constraints.
HbA1c and glycated-protein assays	Minimally invasive specimen	Chromatographic, immunologic or enzymatic measurement of protein glycation.	Average glycemic exposure proxy over different time windows.	Standardized and outcome-linked in many diabetes contexts.	Red-cell lifespan, hemoglobin variants, kidney disease, pregnancy and assay method alter HbA1c; averages hide variability.
Oral glucose and endocrine dynamic tests	Minimally invasive challenge	Standardized substrate or stimulant/suppressant followed by timed sampling.	Reserve, feedback or diagnostic response curves.	Tests physiology not visible at baseline.	Preparation, timing, stress, illness, drugs, absorption and assay differences; thresholds may be probabilistic.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
DXA	Noninvasive low-dose ionizing	Dual-energy X-ray estimates areal bone mineral density and body composition.	BMD T/Z scores, regional lean/fat mass proxies and change.	Standardized fracture-risk component and low burden.	Two-dimensional, machine/analysis dependence, artifacts, body-size effects and limited individual change precision.
Bioimpedance and consumer body composition	Noninvasive electrical	Body impedance at one or more frequencies is mapped to water and composition models.	Total/body water and fat/lean estimates.	Portable and repeatable under controlled conditions.	Hydration, temperature, food, exercise, edema, body geometry and proprietary equations; poor individual inter-changeability.
Indirect calorimetry	Noninvasive respiratory gas	Oxygen consumption and carbon-dioxide production estimate energy expenditure and substrate oxidation.	Resting energy expenditure, respiratory exchange ratio and metabolic response.	Directer physiological estimate than predictive equations.	Leaks, non-steady state, gas calibration, ventilation mode and assumptions about protein metabolism.
Endocrine ultrasound and nuclear imaging	Noninvasive/minimally invasive	Ultrasound anatomy/flow; radioisotope uptake or functional localization.	Thyroid/adrenal/gonadal lesion characterization, uptake and procedure planning.	Anatomic or functional localization after biochemical suspicion.	Incidental nodules, operator dependence, radiation and imaging cannot alone establish hormone activity.
MRI/CT of pituitary and adrenal systems	Noninvasive/minimally invasive imaging	Microscopic anatomy with contrast as needed.	Mass, size, invasion, hemorrhage and procedure planning.	High anatomic detail.	Incidentalomas, protocol dependence, radiation/contrast for CT and anatomy-physiology discordance.
Metabolomics and multi-omics	Minimally invasive specimen/data intensive	High-dimensional molecular profiles of metabolites, genes, transcripts, proteins and epigenetic states.	Pathway hypotheses, subtypes, exposure signatures and candidate biomarkers.	Broad discovery and mechanistic integration.	Batch effects, confounding, multiplicity, unstable signatures, tissue mismatch and weak clinical utility.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Wearable activity, sleep and meal sensing	Noninvasive digital	Inertial, optical, temperature, interaction and logging systems estimate behavior/context.	Activity, sleep timing, heart rate, meal/medication annotations and adherence proxies.	Adds real-world context to glucose and hormone dynamics.	Behavior inference error, missingness, privacy, device changes and no direct measurement of intake or metabolic state.
Bone biopsy and specialized turnover assays	Invasive/minimally invasive	Allystomorphometry or biochemical markers of formation/resorption.	Bone turnover, mineralization and selected renal/metabolic bone classification.	Can distinguish mechanisms not visible on DXA.	Sampling, invasiveness, biologic variability and uncertain individual treatment thresholds.

10.0.1 Clinical history, anthropometry and growth charts

Measured quantity and transducer. Symptoms, medication/exposure history, weight, height, waist, growth velocity and pubertal stage.

Typical outputs. Trajectory, phenotype, nutritional and functional context.

Strengths. Low cost and often more informative than one laboratory value.

Limitations and artifacts. Technique, edema, clothing, body composition, reference population, stigma and recall.

Reference and validation strategy. Standardized equipment/protocol, repeated trajectory and age/life-stage references.

Clinical/research maturity. Mature.

Evidence anchors: ^[8,17]

10.0.2 Laboratory immunoassays

Measured quantity and transducer. Antibody-based detection of hormones, binding proteins, antibodies or peptides.

Typical outputs. Concentration interpreted by timing, reference and axis.

Strengths. Sensitive, scalable and central to endocrine diagnosis.

Limitations and artifacts. Cross-reactivity, heterophile/biotin interference, hook effect, matrix and platform noncomparability.

Reference and validation strategy. Internal/external QC, reference methods, dilution/recovery, alternate platform or mass spectrometry for discordance.

Clinical/research maturity. Mature with assay-specific limits.

Evidence anchors: [7,28]

10.0.3 Mass spectrometry and chromatography

Measured quantity and transducer. Separation and mass/charge detection of steroids, metabolites, amino acids and drugs.

Typical outputs. Specific analyte panels, metabolite patterns and confirmatory measurements.

Strengths. Higher analytical specificity and multiplexing.

Limitations and artifacts. Complex preanalytics, calibration, isomers, cost and laboratory-developed method variability.

Reference and validation strategy. Traceable standards, matrix effects, proficiency testing and clinical reference intervals.

Clinical/research maturity. Mature specialist method.

Evidence anchors: [22,28]

10.0.4 Plasma glucose meter

Measured quantity and transducer. Electrochemical enzymatic strip measures capillary glucose.

Typical outputs. Point glucose for self-management and acute decisions.

Strengths. Fast, portable and actionable.

Limitations and artifacts. Strip/storage, hematocrit, perfusion, contamination, interfering substances and user technique; capillary may differ in shock.

Reference and validation strategy. ISO/FDA performance, lot testing, user studies and comparison across range.

Clinical/research maturity. Mature.

Evidence anchors: [6,23]

10.0.5 Continuous glucose monitoring

Measured quantity and transducer. Enzymatic or optical sensing estimates interstitial glucose continuously.

Typical outputs. Glucose trace, time in range, variability, alerts and automated-delivery input.

Strengths. Reveals dynamics and supports closed-loop control.

Limitations and artifacts. Physiologic lag, compression, calibration/model drift, sensor failure, medication interference, alert burden and access.

Reference and validation strategy. Paired reference across rate/range, wear duration, body sites, subgroup and failure-state testing; outcome trials.

Clinical/research maturity. Mature for diabetes; broader wellness claims limited.

Evidence anchors: [6,23,24]

10.0.6 Insulin pump and automated delivery telemetry

Measured quantity and transducer. Programmed subcutaneous insulin delivery integrated with CGM/control algorithm.

Typical outputs. Delivered dose, basal/bolus history, algorithm state and device alarms.

Strengths. Precise adjustable delivery and reduced burden for selected users.

Limitations and artifacts. Infusion failure can rapidly cause ketoacidosis; cybersecurity, interface, user input and algorithm constraints.

Reference and validation strategy. Delivery accuracy, occlusion/fault testing, software validation, human factors, closed-loop trials and postmarket surveillance.

Clinical/research maturity. Mature regulated system.

Evidence anchors: [6,25,27]

10.0.7 HbA1c and glycated-protein assays

Measured quantity and transducer. Chromatographic, immunologic or enzymatic measurement of protein glycation.

Typical outputs. Average glycemic exposure proxy over different time windows.

Strengths. Standardized and outcome-linked in many diabetes contexts.

Limitations and artifacts. Red-cell lifespan, hemoglobin variants, kidney disease, pregnancy and assay method alter HbA1c; averages hide variability.

Reference and validation strategy. Traceability, variant/interference testing and comparison with glucose patterns.

Clinical/research maturity. Mature.

Evidence anchors: [6,11]

10.0.8 Oral glucose and endocrine dynamic tests

Measured quantity and transducer. Standardized substrate or stimulant/suppressant followed by timed sampling.

Typical outputs. Reserve, feedback or diagnostic response curves.

Strengths. Tests physiology not visible at baseline.

Limitations and artifacts. Preparation, timing, stress, illness, drugs, absorption and assay differences; thresholds may be probabilistic.

Reference and validation strategy. Protocol fidelity, timing verification, reference population and outcome linkage.

Clinical/research maturity. Mature for defined axes.

Evidence anchors: [6,7]

10.0.9 DXA

Measured quantity and transducer. Dual-energy X-ray estimates areal bone mineral density and body composition.

Typical outputs. BMD T/Z scores, regional lean/fat mass proxies and change.

Strengths. Standardized fracture-risk component and low burden.

Limitations and artifacts. Two-dimensional, machine/analysis dependence, artifacts, body-size effects and limited individual change precision.

Reference and validation strategy. Phantom QA, cross-calibration, least significant change and trained positioning.

Clinical/research maturity. Mature.

Evidence anchors: [9,28]

10.0.10 Bioimpedance and consumer body composition

Measured quantity and transducer. Body impedance at one or more frequencies is mapped to water and composition models.

Typical outputs. Total/body water and fat/lean estimates.

Strengths. Portable and repeatable under controlled conditions.

Limitations and artifacts. Hydration, temperature, food, exercise, edema, body geometry and proprietary equations; poor individual interchangeability.

Reference and validation strategy. Electrical phantoms, standardized state, reference imaging/dilution and subgroup validation.

Clinical/research maturity. Context-limited; consumer trends require caution.

Evidence anchors: ^[23,42]

10.0.11 Indirect calorimetry

Measured quantity and transducer. Oxygen consumption and carbon-dioxide production estimate energy expenditure and substrate oxidation.

Typical outputs. Resting energy expenditure, respiratory exchange ratio and metabolic response.

Strengths. Directer physiological estimate than predictive equations.

Limitations and artifacts. Leaks, non-steady state, gas calibration, ventilation mode and assumptions about protein metabolism.

Reference and validation strategy. Gas/flow calibration, steady-state criteria, repeated measures and reference combustion/metabolic methods.

Clinical/research maturity. Mature specialized method.

Evidence anchors: ^[4]

10.0.12 Endocrine ultrasound and nuclear imaging

Measured quantity and transducer. Ultrasound anatomy/flow; radioisotope uptake or functional localization.

Typical outputs. Thyroid/adrenal/gonadal lesion characterization, uptake and procedure planning.

Strengths. Anatomic or functional localization after biochemical suspicion.

Limitations and artifacts. Incidental nodules, operator dependence, radiation and imaging cannot alone establish hormone activity.

Reference and validation strategy. Phantom/reader reliability and pathology/biochemical correlation.

Clinical/research maturity. Mature for defined indications.

Evidence anchors: [7,31]

10.0.13 MRI/CT of pituitary and adrenal systems

Measured quantity and transducer. Cross-sectional anatomy with contrast as needed.

Typical outputs. Mass, size, invasion, hemorrhage and procedure planning.

Strengths. High anatomic detail.

Limitations and artifacts. Incidentalomas, protocol dependence, radiation/contrast for CT and anatomy-physiology discordance.

Reference and validation strategy. Protocol QA, blinded reads, longitudinal and surgical/pathologic reference.

Clinical/research maturity. Mature.

Evidence anchors: [7,42]

10.0.14 Metabolomics and multi-omics

Measured quantity and transducer. High-dimensional molecular profiles of metabolites, genes, transcripts, proteins and epigenetic states.

Typical outputs. Pathway hypotheses, subtypes, exposure signatures and candidate biomarkers.

Strengths. Broad discovery and mechanistic integration.

Limitations and artifacts. Batch effects, confounding, multiplicity, unstable signatures, tissue mismatch and weak clinical utility.

Reference and validation strategy. Standards, randomized sample order, external replication, causal triangulation and utility trials.

Clinical/research maturity. Research/frontier for most individual claims.

Evidence anchors: [22,42]

10.0.15 Wearable activity, sleep and meal sensing

Measured quantity and transducer. Inertial, optical, temperature, interaction and logging systems estimate behavior/context.

Typical outputs. Activity, sleep timing, heart rate, meal/medication annotations and adherence proxies.

Strengths. Adds real-world context to glucose and hormone dynamics.

Limitations and artifacts. Behavior inference error, missingness, privacy, device changes and no direct measurement of intake or metabolic state.

Reference and validation strategy. Reference observation/calorimetry/sleep testing, free-living validation and version tracking.

Clinical/research maturity. Established narrow metrics; composite metabolic scores frontier.

Evidence anchors: ^[19,23]

10.0.16 Bone biopsy and specialized turnover assays

Measured quantity and transducer. Histomorphometry or biochemical markers of formation/resorption.

Typical outputs. Bone turnover, mineralization and selected renal/metabolic bone classification.

Strengths. Can distinguish mechanisms not visible on DXA.

Limitations and artifacts. Sampling, invasiveness, biologic variability and uncertain individual treatment thresholds.

Reference and validation strategy. Specialist protocols, assay standardization and clinical/pathologic correlation.

Clinical/research maturity. Specialized.

Evidence anchors: ^[9,11]

Engineering architectures, signal quality, and validation

Buildable systems should begin with an explicit intended use, measurand, reference method, population, environment, and failure policy. Prototype plausibility is not clinical validity. Raw data, quality channels, timing, calibration state, firmware version, and provenance should be retained so that errors can be reconstructed rather than hidden by a polished score.

Subsystem / design issue	Requirements	Dominant risks	Verification
Intended use and measurand	Define the precise endocrine and metabolic health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.	A broad wellness claim can hide several incompatible clinical claims.	Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.
Sensor and transduction	Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.	A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.	Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.
Timing and synchronization	Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.	Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.	Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.
Analog, acquisition, and digital integrity	Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.	Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.	Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.
Calibration and traceability	Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.	Model calibration to one device lot, site, or operator may not transport.	Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.
Artifact and quality channels	Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.	Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.	Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

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Subsystem / design issue	Requirements	Dominant risks	Verification
Algorithms and uncertainty	Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.	Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.	Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.
Safety, usability, cybersecurity, and privacy	Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.	Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.	Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

11.0.1 Intended use and measurand

Design requirements. Define the precise endocrine and metabolic health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.

Failure modes. A broad wellness claim can hide several incompatible clinical claims.

Bench verification. Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.

Human validation boundary. Do not enroll people until the measurement purpose and risk controls are reviewable.

Evidence anchors: [23,29]

11.0.2 Sensor and transduction

Design requirements. Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.

Failure modes. A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.

Bench verification. Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.

Human validation boundary. Bench testing should precede contact with participants.

Evidence anchors: [29]

11.0.3 Timing and synchronization

Design requirements. Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.

Failure modes. Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.

Bench verification. Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.

Human validation boundary. Store both device time and host/reference time.

11.0.4 Analog, acquisition, and digital integrity

Design requirements. Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.

Failure modes. Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.

Bench verification. Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.

Human validation boundary. Preserve raw or minimally processed channels where ethically and technically possible.

Evidence anchors: ^[27,28]

11.0.5 Calibration and traceability

Design requirements. Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.

Failure modes. Model calibration to one device lot, site, or operator may not transport.

Bench verification. Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.

Human validation boundary. Calibration data and version must travel with each observation.

Evidence anchors: ^[28]

11.0.6 Artifact and quality channels

Design requirements. Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.

Failure modes. Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.

Bench verification. Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Human validation boundary. Quality rejection must not silently exclude populations.

Evidence anchors: ^[42]

11.0.7 Algorithms and uncertainty

Design requirements. Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.

Failure modes. Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.

Bench verification. Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.

Human validation boundary. A model should abstain when inputs are outside validated conditions.

Evidence anchors: ^[24,42]

11.0.8 Safety, usability, cybersecurity, and privacy

Design requirements. Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.

Failure modes. Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.

Bench verification. Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

Human validation boundary. Research exemptions do not remove the ethical duty to minimize foreseeable harm.

Evidence anchors: ^[25]

Human-contact engineering rule

Do not connect a human participant to a mains-referenced oscilloscope, programmer, charger, or non-medical ground path. Bench characterization should use simulators and isolated fixtures. Human research requires documented electrical safety, risk management, consent, privacy controls, and appropriate oversight.

Prevention, treatment, rehabilitation, and system intervention

Interventions are classified by target and level: upstream determinants, exposure reduction, behavioral support, prevention, diagnosis-linked treatment, acute rescue, chronic management, rehabilitation, assistive technology, and palliative or supportive care. Net benefit depends on baseline risk, absolute effect, burden, interactions, access, adherence, and the outcomes valued by the person.

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Food-system and environmental prevention	Population/upstream	Improve access to affordable nutritious food, safe activity, sleep-compatible work, education and reduced marketing/exposure drivers.	Population evidence supports structural determinants; isolated individual advice has limited reach.	Unintended substitution, inequitable implementation and political/economic barriers.	Food security, affordability, dietary patterns, activity opportunity, metabolic outcomes and equity.
Structured nutrition, activity and behavior support	Behavioral/clinical	Create sustainable energy, nutrient, meal-timing and activity patterns aligned with disease, culture, resources and goals.	Effective for diabetes prevention/control, weight and function when sufficiently intensive and supported.	Stigma, eating disorder risk, injury, cost, time and rebound after program withdrawal.	Glucose/weight/fat, diet quality, strength, symptoms, mental health and sustainability.
Insulin replacement and closed-loop delivery	Hormone/device/therapy	Replace basal and prandial insulin and adapt to glucose, meals, activity and illness.	Life-saving in deficiency; CGM and automated delivery improve time in range and reduce selected hypoglycemia burden.	Hypoglycemia, weight gain, infusion/device failure, data burden, cost and algorithm limits.	CGM metrics plus symptoms, severe events, HbA1c, dose, sites, alarms and quality of life.
Non-insulin glucose-lowering pharmacotherapy	Pharmacology	Modify insulin sensitivity/secretion, hepatic output, incretin signaling, renal glucose handling or absorption.	Benefits vary; several classes reduce cardiovascular/kidney events or weight in indicated populations beyond glucose lowering.	GI effects, hypoglycemia for selected agents, genital infection, ketoacidosis, fluid effects, cost and interactions.	Glucose, renal/hepatic function, weight, cardiorenal outcomes, adverse effects and adherence.

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Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Anti-obesity pharmacotherapy	Pharmacology	Alter appetite, satiety, gut-brain signaling or nutrient handling to support sustained weight and complication reduction.	Current incretin-based and other therapies can produce clinically substantial weight loss and improve several risk factors/outcomes in selected groups.	GI and gallbladder effects, lean-mass loss, rare serious events, supply/cost, discontinuation regain and pregnancy restrictions.	Weight/body composition and function, complications, nutrition, adverse effects, mental health and continuation criteria.
Metabolic/bariatric surgery	Procedure	Alter GI anatomy and signaling to produce durable weight and metabolic effects.	Most effective durable intervention for severe obesity in selected candidates; can improve diabetes and mortality-related outcomes.	Operative risk, micronutrient deficiency, hypoglycemia, alcohol/substance shifts, reoperation and access inequity.	Nutrition/labs, weight and diabetes, complications, bone health, mental health and long-term follow-up.
Hormone replacement	Pharmacology	Replace thyroid, adrenal, pituitary, gonadal or other hormones at physiologic goals.	Essential for true deficiency; dosing and formulation are axis- and life-stage-specific.	Over/underreplacement, interactions, adherence, circadian mismatch and monitoring burden.	Clinical response, timed labs, organ effects, bone/cardiovascular risk and emergency education where relevant.
Hormone suppression or receptor blockade	Pharmacology	Reduce synthesis, secretion, conversion or receptor effect in hormone excess.	Effective for many thyroid, adrenal, pituitary and androgen/estrogen disorders as bridge or long-term therapy.	Organ toxicity, reproductive effects, deficiency, interactions and incomplete tumor control.	Axis-specific hormones, symptoms, organ toxicity, imaging/tumor status and replacement needs.
Endocrine surgery and ablation	Procedure	Remove or ablate hormone-producing or compressive tissue.	Can cure selected thyroid, parathyroid, adrenal or pituitary disease.	Bleeding, nerve/organ injury, permanent hormone deficiency, recurrence and incidental overtreatment.	Biochemical cure, pathology, deficits, imaging and long-term recurrence.
Bone-active therapy and fall prevention	Pharmacology	Reduce resorption, stimulate formation, improve strength/balance and remove secondary causes.	Reduces fractures in appropriately selected high-risk people; sequence and adherence matter.	Rare jaw/femur events, hypocalcemia, rebound, injection burden and falls from other causes.	Fractures, DXA with least significant change, calcium/vitamin D/renal context, falls and adherence.

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Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Rare-disease substrate, enzyme and gene/RNA therapy	Precision/advanced therapy	Correct a defined biochemical or genetic defect through replacement, reduction or molecular modification.	Transformative for a growing but small set of conditions; effects vary by organ damage and timing.	Immune reactions, off-target/long-term uncertainty, irreversible risk, extreme cost and access.	Biochemical target, organ function, immunogenicity, adverse events, durability and patient-important function.
Multidisciplinary and psychosocial support	Care system	Integrate diabetes/endocrine expertise, nutrition, pharmacy, mental health, reproductive, social and primary care.	Improves safety and self-management for complex chronic disease when coordinated.	Visit and data burden, conflicting advice, stigma and unequal access.	Goal attainment, treatment burden, distress, severe events, access and continuity.

12.0.1 Food-system and environmental prevention

Target and mechanism. Improve access to affordable nutritious food, safe activity, sleep-compatible work, education and reduced marketing/exposure drivers.

Evidence and expected outcome. Population evidence supports structural determinants; isolated individual advice has limited reach.

Risks, burden, and equity. Unintended substitution, inequitable implementation and political/economic barriers.

Response monitoring. Food security, affordability, dietary patterns, activity opportunity, metabolic outcomes and equity.

Evidence anchors: [16,34,36]

12.0.2 Structured nutrition, activity and behavior support

Target and mechanism. Create sustainable energy, nutrient, meal-timing and activity patterns aligned with disease, culture, resources and goals.

Evidence and expected outcome. Effective for diabetes prevention/control, weight and function when sufficiently intensive and supported.

Risks, burden, and equity. Stigma, eating disorder risk, injury, cost, time and rebound after program withdrawal.

Response monitoring. Glucose/weight/function, diet quality, strength, symptoms, mental health and sustainability.

Evidence anchors: [6,8,10]

12.0.3 Insulin replacement and closed-loop delivery

Target and mechanism. Replace basal and prandial insulin and adapt to glucose, meals, activity and illness.

Evidence and expected outcome. Life-saving in deficiency; CGM and automated delivery improve time in range and reduce selected hypoglycemia burden.

Risks, burden, and equity. Hypoglycemia, weight gain, infusion/device failure, data burden, cost and algorithm limits.

Response monitoring. CGM metrics plus symptoms, severe events, HbA1c, dose, sites, alarms and quality of life.

Evidence anchors: [6,24,25]

12.0.4 Non-insulin glucose-lowering pharmacotherapy

Target and mechanism. Modify insulin sensitivity/secretion, hepatic output, incretin signaling, renal glucose handling or absorption.

Evidence and expected outcome. Benefits vary; several classes reduce cardiovascular/kidney events or weight in indicated populations beyond glucose lowering.

Risks, burden, and equity. GI effects, hypoglycemia for selected agents, genital infection, ketoacidosis, fluid effects, cost and interactions.

Response monitoring. Glucose, renal/hepatic function, weight, cardiorenal outcomes, adverse effects and adherence.

Evidence anchors: [2,6,11]

12.0.5 Anti-obesity pharmacotherapy

Target and mechanism. Alter appetite, satiety, gut-brain signaling or nutrient handling to support sustained weight and complication reduction.

Evidence and expected outcome. Current incretin-based and other therapies can produce clinically substantial weight loss and improve several risk factors/outcomes in selected groups.

Risks, burden, and equity. GI and gallbladder effects, lean-mass loss, rare serious events, supply/cost, discontinuation regain and pregnancy restrictions.

Response monitoring. Weight/body composition and function, complications, nutrition, adverse effects, mental health and continuation criteria.

Evidence anchors: [6,8,26]

12.0.6 Metabolic/bariatric surgery

Target and mechanism. Alter GI anatomy and signaling to produce durable weight and metabolic effects.

Evidence and expected outcome. Most effective durable intervention for severe obesity in selected candidates; can improve diabetes and mortality-related outcomes.

Risks, burden, and equity. Operative risk, micronutrient deficiency, hypoglycemia, alcohol/substance shifts, reoperation and access inequity.

Response monitoring. Nutrition/labs, weight and diabetes, complications, bone health, mental health and long-term follow-up.

Evidence anchors: [8,39,40]

12.0.7 Hormone replacement

Target and mechanism. Replace thyroid, adrenal, pituitary, gonadal or other hormones at physiologic goals.

Evidence and expected outcome. Essential for true deficiency; dosing and formulation are axis- and life-stage-specific.

Risks, burden, and equity. Over/underreplacement, interactions, adherence, circadian mismatch and monitoring burden.

Response monitoring. Clinical response, timed labs, organ effects, bone/cardiovascular risk and emergency education where relevant.

Evidence anchors: [7,30]

12.0.8 Hormone suppression or receptor blockade

Target and mechanism. Reduce synthesis, secretion, conversion or receptor effect in hormone excess.

Evidence and expected outcome. Effective for many thyroid, adrenal, pituitary and androgen/estrogen disorders as bridge or long-term therapy.

Risks, burden, and equity. Organ toxicity, reproductive effects, deficiency, interactions and incomplete tumor control.

Response monitoring. Axis-specific hormones, symptoms, organ toxicity, imaging/tumor status and replacement needs.

Evidence anchors: [7,26]

12.0.9 Endocrine surgery and ablation

Target and mechanism. Remove or ablate hormone-producing or compressive tissue.

Evidence and expected outcome. Can cure selected thyroid, parathyroid, adrenal or pituitary disease.

Risks, burden, and equity. Bleeding, nerve/organ injury, permanent hormone deficiency, recurrence and incidental overtreatment.

Response monitoring. Biochemical cure, pathology, deficits, imaging and long-term recurrence.

Evidence anchors: [7,31]

12.0.10 Bone-active therapy and fall prevention

Target and mechanism. Reduce resorption, stimulate formation, improve strength/balance and remove secondary causes.

Evidence and expected outcome. Reduces fractures in appropriately selected high-risk people; sequence and adherence matter.

Risks, burden, and equity. Rare jaw/femur events, hypocalcemia, rebound, injection burden and falls from other causes.

Response monitoring. Fractures, DXA with least significant change, calcium/vitamin D/renal context, falls and adherence.

Evidence anchors: [9,33]

12.0.11 Rare-disease substrate, enzyme and gene/RNA therapy

Target and mechanism. Correct a defined biochemical or genetic defect through replacement, reduction or molecular modification.

Evidence and expected outcome. Transformative for a growing but small set of conditions; effects vary by organ damage and timing.

Risks, burden, and equity. Immune reactions, off-target/long-term uncertainty, irreversible risk, extreme cost and access.

Response monitoring. Biochemical target, organ function, immunogenicity, adverse events, durability and patient-important function.

Evidence anchors: [21,24,32]

12.0.12 Multidisciplinary and psychosocial support

Target and mechanism. Integrate diabetes/endocrine expertise, nutrition, pharmacy, mental health, reproductive, social and primary care.

Evidence and expected outcome. Improves safety and self-management for complex chronic disease when coordinated.

Risks, burden, and equity. Visit and data burden, conflicting advice, stigma and unequal access.

Response monitoring. Goal attainment, treatment burden, distress, severe events, access and continuity.

Evidence anchors: ^[34,35,38]

What is known, not known, and frequently overstated

Uncertainty is not a single residual. It can arise from biological variability, measurement error, selection, confounding, limited follow-up, inadequate reference standards, endpoint substitution, poor external validity, causal ambiguity, or value disagreement. The correct response is to identify the uncertainty and design the next measurement or study.

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
How should obesity be defined beyond BMI?	BMI is useful for population screening and correlates with risk, while complications and fat distribution improve individual context.	A globally transferable diagnostic framework that separates adiposity, function and organ risk without increasing stigma or overtreatment.	Body composition methods differ; risk varies by age and population, and thresholds interact with access to treatment.	Prospective diverse cohorts comparing simple and advanced phenotypes against patient-important outcomes and treatment response.
Can continuous multi-hormone monitoring become clinically useful?	Glucose monitoring shows the value of dynamics; several hormone sensors are technically plausible.	Accuracy, stability, biocompatibility and decision utility for cortisol, lactate, insulin or other analytes in daily life.	Low concentrations, binding, biofouling, interstitial kinetics, circadian variation and uncertain actions challenge sensing.	Bench metrology, microdialysis/reference studies and narrowly defined utility trials.
What is the long-term optimal use of potent weight-loss medication?	Continued treatment often maintains effect while discontinuation commonly leads to regain.	Best duration, sequencing, maintenance dosing, lean-mass protection and population-level value.	Trials have limited duration and selected participants; access and adherence alter real-world trajectories.	Long pragmatic comparative trials with body composition, function, events, discontinuation and equity outcomes.
Can endocrine multi-omics define actionable subtypes?	Molecular profiles reveal heterogeneity and pathways.	Whether subtypes are stable, causal and improve treatment over clinical variables.	High-dimensional batch effects and tissue mismatch produce nontransportable clusters.	Prospective locked classifier trials with treatment interaction and clinical utility endpoints.

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Question	What is established	What remains uncertain	Why unresolved	Decisive next step
How much risk is genetic versus environmental?	Both inherited variation and environments contribute, often through interaction and mediation across the life course.	No single percentage applies to all outcomes, populations, ages, or definitions within endocrine and metabolic health.	Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.	Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.
Can one composite score summarize health?	Composite scores can rank a defined outcome in a defined population and may improve workflow.	Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.	Different variables compensate, interact, and have different treatment implications; weights drift over time.	Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.
When is earlier detection beneficial?	Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.	Net benefit for many incidental, molecular, wearable, and multi-cancer signals.	Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.	Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.
Do digital biomarkers generalize?	Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.	Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.	Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.	Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.
What counts as meaningful change?	Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.	Thresholds for many biomarkers, sensors, and multidomain profiles.	Baseline severity, goals, adaptation, and treatment burden alter value.	Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

13.0.1 How should obesity be defined beyond BMI?

Established knowledge. BMI is useful for population screening and correlates with risk, while complications and fat distribution improve individual context.

Unresolved issue. A globally transferable diagnostic framework that separates adiposity, function and organ risk without increasing stigma or overtreatment.

Why current evidence is insufficient. Body composition methods differ, risk varies by age and population, and thresholds interact with access to treatment.

Research that would change confidence. Prospective diverse cohorts comparing simple and advanced phenotypes against patient-important outcomes and treatment response.

Evidence anchors: [8,34]

13.0.2 Can continuous multi-hormone monitoring become clinically useful?

Established knowledge. Glucose monitoring shows the value of dynamics; several hormone sensors are technically plausible.

Unresolved issue. Accuracy, stability, biocompatibility and decision utility for cortisol, lactate, insulin or other analytes in daily life.

Why current evidence is insufficient. Low concentrations, binding, biofouling, interstitial kinetics, circadian variation and uncertain actions challenge sensing.

Research that would change confidence. Bench metrology, microdialysis/reference studies and narrowly defined utility trials.

Evidence anchors: [23,29]

13.0.3 What is the long-term optimal use of potent weight-loss medication?

Established knowledge. Continued treatment often maintains effect while discontinuation commonly leads to regain.

Unresolved issue. Best duration, sequencing, maintenance dosing, lean-mass protection and population-level value.

Why current evidence is insufficient. Trials have limited duration and selected participants; access and adherence alter real-world trajectories.

Research that would change confidence. Long pragmatic comparative trials with body composition, function, events, discontinuation and equity outcomes.

Evidence anchors: [8,41]

13.0.4 Can endocrine multi-omics define actionable subtypes?

Established knowledge. Molecular profiles reveal heterogeneity and pathways.

Unresolved issue. Whether subtypes are stable, causal and improve treatment over clinical variables.

Why current evidence is insufficient. High-dimensional batch effects and tissue mismatch produce non-transportable clusters.

Research that would change confidence. Prospective locked classifier trials with treatment interaction and clinical utility endpoints.

Evidence anchors: [22,42]

13.0.5 How much risk is genetic versus environmental?

Established knowledge. Both inherited variation and environments contribute, often through interaction and mediation across the life course.

Unresolved issue. No single percentage applies to all outcomes, populations, ages, or definitions within endocrine and metabolic health.

Why current evidence is insufficient. Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.

Research that would change confidence. Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

Evidence anchors: [21]

13.0.6 Can one composite score summarize health?

Established knowledge. Composite scores can rank a defined outcome in a defined population and may improve workflow.

Unresolved issue. Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.

Why current evidence is insufficient. Different variables compensate, interact, and have different treatment implications; weights drift over time.

Research that would change confidence. Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Evidence anchors: [42]

13.0.7 When is earlier detection beneficial?

Established knowledge. Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.

Unresolved issue. Net benefit for many incidental, molecular, wearable, and multi-cancer signals.

Why current evidence is insufficient. Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.

Research that would change confidence. Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.

Evidence anchors: ^[41]

13.0.8 Do digital biomarkers generalize?

Established knowledge. Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.

Unresolved issue. Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.

Why current evidence is insufficient. Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.

Research that would change confidence. Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.

Evidence anchors: ^[23]

13.0.9 What counts as meaningful change?

Established knowledge. Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.

Unresolved issue. Thresholds for many biomarkers, sensors, and multidomain profiles.

Why current evidence is insufficient. Baseline severity, goals, adaptation, and treatment burden alter value.

Research that would change confidence. Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

Research and development roadmap

The roadmap prioritizes work that changes decisions or reduces uncertainty, not merely work that generates more variables. Each project should declare the causal or measurement target, comparator, reference, population, sampling frame, missing-data plan, subgroup analysis, and stopping criteria before data collection.

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
0. Assay harmonization atlas	How much diagnostic disagreement is caused by platform, binding proteins and preanalytics?	Split-sample multi-platform study across pregnancy, kidney/liver disease, acute illness, supplements and interference-positive specimens.	Traceable conversion/abstention rules reduce clinically consequential disagreement.	Differences remain unexplained or corrections only work in one laboratory.
0. Closed-loop failure registry	Which sensor, algorithm, infusion and interface failures cause severe hypoglycemia or ketoacidosis?	Prospective device-version registry with raw logs, root-cause analysis, human factors and denominator exposure.	Specific design and alarm changes measurably reduce severe events.	Only voluntary narratives without device/firmware denominators.
0. Metabolic health utility trial	Does a multidomain profile outperform single-disease care without increasing burden?	Pragmatic cluster trial integrating glucose, kidney, liver, cardiovascular, sleep, function and social support.	Improved patient-important outcomes, treatment burden and equity at acceptable cost.	More tests and medication with no net benefit.
1. Reference map	What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for endocrine and metabolic health?	Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.	Reproducible distributions and state-dependent reference models with quantified uncertainty.	Convenience sample presented as universal normality.
2. Causal determinants	Which modifiable exposures cause clinically and functionally important outcomes, and in whom?	Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.	Convergent effect estimates linked to plausible mechanisms and actionable interventions.	Association-only result marketed as prevention guidance.
3. Measurement validity	Does the new device or biomarker measure the intended quantity under real use?	Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.	Prespecified error and failure limits met across intended-use conditions.	High average correlation with systematic individual error or unreported exclusions.

Continued on next page

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
4. Clinical utility	Does using the measurement improve decisions or outcomes compared with current care?	Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.	Net benefit on patient-important outcomes with acceptable implementation burden.	Improved prediction without changed decisions or outcomes.
5. Longitudinal multimorbidity	How does the domain interact with other conditions, treatments, and functional trajectories?	Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.	Models that support coordinated care rather than isolated disease optimization.	Single-disease endpoint worsens overall burden or conflicts with goals.
6. Implementation and governance	Can effective interventions be delivered safely, equitably, affordably, and sustainably?	Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.	Sustained adoption with monitored outcomes and reduced inequity.	Pilot success depends on exceptional staffing or hidden externalized costs.

14.0.1 0. Assay harmonization atlas

Question. How much diagnostic disagreement is caused by platform, binding proteins and preanalytics?

Design. Split-sample multi-platform study across pregnancy, kidney/liver disease, acute illness, supplements and interference-positive specimens.

Success criterion. Traceable conversion/abstention rules reduce clinically consequential disagreement.

Failure or stop criterion. Differences remain unexplained or corrections only work in one laboratory.

Evidence anchors: [7,28]

14.0.2 0. Closed-loop failure registry

Question. Which sensor, algorithm, infusion and interface failures cause severe hypoglycemia or ketoacidosis?

Design. Prospective device-version registry with raw logs, root-cause analysis, human factors and denominator exposure.

Success criterion. Specific design and alarm changes measurably reduce severe events.

Failure or stop criterion. Only voluntary narratives without device/firmware denominators.

Evidence anchors: [25,31]

14.0.3 0. Metabolic health utility trial

Question. Does a multidomain profile outperform single-disease care without increasing burden?

Design. Pragmatic cluster trial integrating glucose, kidney, liver, cardiovascular, sleep, function and social support.

Success criterion. Improved patient-important outcomes, treatment burden and equity at acceptable cost.

Failure or stop criterion. More tests and medication with no net benefit.

Evidence anchors: [6,34,37]

14.0.4 1. Reference map

Question. What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for endocrine and metabolic health?

Design. Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.

Success criterion. Reproducible distributions and state-dependent reference models with quantified uncertainty.

Failure or stop criterion. Convenience sample presented as universal normality.

Evidence anchors: [34]

14.0.5 2. Causal determinants

Question. Which modifiable exposures cause clinically and functionally important outcomes, and in whom?

Design. Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.

Success criterion. Convergent effect estimates linked to plausible mechanisms and actionable interventions.

Failure or stop criterion. Association-only result marketed as prevention guidance.

14.0.6 3. Measurement validity

Question. Does the new device or biomarker measure the intended quantity under real use?

Design. Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.

Success criterion. Prespecified error and failure limits met across intended-use conditions.

Failure or stop criterion. High average correlation with systematic individual error or unreported exclusions.

Evidence anchors: [29,42]

14.0.7 4. Clinical utility

Question. Does using the measurement improve decisions or outcomes compared with current care?

Design. Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.

Success criterion. Net benefit on patient-important outcomes with acceptable implementation burden.

Failure or stop criterion. Improved prediction without changed decisions or outcomes.

Evidence anchors: [37,41]

14.0.8 5. Longitudinal multimorbidity

Question. How does the domain interact with other conditions, treatments, and functional trajectories?

Design. Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.

Success criterion. Models that support coordinated care rather than isolated disease optimization.

Failure or stop criterion. Single-disease endpoint worsens overall burden or conflicts with goals.

14.0.9 6. Implementation and governance

Question. Can effective interventions be delivered safely, equitably, affordably, and sustainably?

Design. Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.

Success criterion. Sustained adoption with monitored outcomes and reduced inequity.

Failure or stop criterion. Pilot success depends on exceptional staffing or hidden externalized costs.

Evidence anchors: [34,37]

Cross-volume integration

The health atlas is deliberately polyhierarchical. A phenomenon can have one primary home while also participating in multiple causal and functional systems. Cross-references below identify the most important companion volumes and the reason for the link.

Volume	Companion topic	Relationship	Use this link when...
00	Atlas of Human Health	Master taxonomy, history of classification, and multi-axial architecture.	locating the primary home of a concept or mapping classifications.
13	Measurement, Evidence, Data, and Informatics	Shared metrology, causal inference, validation, interoperability, and data governance.	designing a study, sensor, dataset, model, or evidence claim.
14	Interventions and Therapeutics	Cross-domain pharmacology, procedures, behavioral interventions, rehabilitation, implementation, and safety.	moving from measurement to action.
15	Ethics, Law, Rights, Equity, and Governance	Consent, privacy, fairness, rights, research oversight, and distribution of benefits and harms.	people, identifiable data, invasive procedures, or consequential algorithms are involved.
03A	Cardiovascular Health	Blood pressure, atherogenic particles, vascular complications and cardiorenal-metabolic treatment effects.	metabolic biomarkers are interpreted as cardiovascular risk or treatment targets.
03E	Digestive, Hepatobiliary and Pancreatic Health	Nutrient absorption, incretins, fatty liver, bile and exocrine pancreatic function.	food intake or metabolism produces GI/liver/pancreatic disease.
03F	Renal and Urinary Health	Glucose drugs, electrolyte/water balance, diabetic kidney disease and mineral metabolism.	filtration changes assays, dosing or hormone interpretation.
03I	Musculoskeletal and Connective Tissue Health	Muscle, bone, loading, frailty and body composition.	weight or endocrine disease affects function and fracture.
03M	Reproductive and Sexual Health	Gonadal axes, fertility, PCOS and sexual function.	endocrine phenotype is expressed through reproduction.
03N	Maternal, Fetal and Perinatal Health	Gestational diabetes, thyroid/adrenal changes and developmental programming.	pregnancy or early life changes reference and risk.
11	Population Health and Prevention	Food systems, obesity/diabetes prevention and policy.	individual counseling is insufficient without structural action.

Cross-reference rule

Reference the primary home for the full mechanism or standard; restate only what is necessary for the local argument. When a claim spans domains, record the bridge explicitly rather than assigning the entire phenomenon to one specialty.

Minimum data schema for a longitudinal domain profile

Layer	Minimum fields	Critical caution
Identity and context	Participant identifier, consent, demographics relevant to the question, setting, language, access, and social/environmental context.	Avoid collecting identity variables without purpose; preserve privacy and provenance.
Time	Absolute timestamps, timezone, age/life stage, episode boundaries, event markers, medication and exposure timing.	Clock drift and ambiguous time zones can invalidate multimodal inference.
Raw signal / specimen	Unprocessed samples, units, sensor channel, specimen type, collection protocol, calibration and storage state.	Derived scores without raw or quality data are difficult to audit.
Quality	Contact, motion, saturation, missingness, assay controls, operator notes, confidence, out-of-range flags.	Quality must be a first-class variable, not only an exclusion after analysis.
Reference	Comparator method, instrument, version, observer protocol, laboratory traceability, adjudication.	A weak reference caps the validity of the new system.
Phenotype and function	Symptoms, signs, diagnoses, severity, capability, participation, patient-reported outcomes, goals.	Disease codes alone do not represent lived outcome.
Intervention and response	Dose, route, device settings, behavior support, adherence, co-interventions, adverse events, response window.	Confounding by indication and treatment changes must be modeled.
Governance	Protocol version, data-use agreement, access log, retention, deletion, model version, incident record.	Data stewardship is part of scientific validity.

Glossary

Term	Operational meaning in this volume
Counterregulation	Hormonal and neural responses that oppose a falling glucose or other threatened controlled variable.
Dynamic test	A standardized stimulus or suppression protocol used to reveal endocrine reserve or feedback.
Ectopic fat	Lipid accumulation in organs not specialized for safe long-term storage, such as liver, muscle or pancreas.
Endocrine axis	Hierarchical and feedback-linked organs and signals regulating a hormone system.
Insulin resistance	Reduced biological response to a given insulin exposure, varying by tissue and state.
Time in range	Proportion of CGM readings within a prespecified glucose interval; interpretation requires data sufficiency and context.
Allostasis	Adaptive regulation that changes physiological activity to meet anticipated or actual demands; chronic or repeated activation can carry cost.
Biomarker	Defined characteristic measured as an indicator of normal biology, pathogenic process, exposure, or response; not automatically a surrogate endpoint.
Function	What a person can do in body functions, activities, and participation, interpreted in context.
Homeostasis	Regulation of variables within viable ranges through feedback, feedforward, redundancy, and adaptation.
Phenotype	Observable or measurable characteristics arising from biology, development, environment, behavior, context, and measurement conditions.
Risk factor	Characteristic associated with future outcome; causal status and modifiability must be established separately.
Calibration	Establishing the relationship between an instrument output and a reference quantity under specified conditions.
Clinical validity	Degree to which a measurement or model is associated with or predicts the clinical state or outcome it claims to represent.
Intended use	The specific purpose, population, setting, user, input, and decision for which a device or model is designed.
Reference method	Comparator selected to estimate truth for validation; every reference has its own error and scope.
Reserve	Capacity to maintain function under challenge beyond resting baseline.
Trajectory	Direction, rate, variability, and reversibility of change over time.

Selected references and current standards

The bibliography emphasizes official standards, current guidelines, landmark primary work, and major syntheses. Inclusion does not imply that every recommendation is universal or that institutional endorsement removes uncertainty. Current online materials were checked through the stated evidence cutoff.

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Version note

Foundation edition generated August 2026. Evidence, guidance, and standards cutoff: 13 August 2026. The suite should be versioned when major classifications, guidelines, device standards, or decisive evidence materially change the map.